D. Training Hardware and GPU hours

We additionally report the number of accelerators used to train each *Pythia* model size, alongside counts of total GPU-hours required for training our models at the throughputs that we achieve.

Model Size	GPU Count	Total GPU hours required
70 M	32	510
160 M	32	1,030
410 M	32	2,540
1.0 B	64	4,830
1.4 B	64	7,120
2.8 B	64	14,240
6.9 B	128	33,500
12 B	256	72,300
Total		136,070

Table 5. Model sizes in the *Pythia* suite, number of GPUs used during training, and the total number of GPU hours, calculated via (iteration time (s) $\times$ number of iterations $\times$ number of GPUs $\div$ 3600 s/hour). All GPUs are A100s with 40GB of memory.

Here "total" refers to training one model of each size in our suite. For this paper, we trained two models of each size (one on the Pile and one on the Pile deduplicated) and had to retrain both model suites an additional time as discussed in Appendix B. Thus the total compute required for training the models for this paper was 544,280 A100-hours.